NAYEEM SARKER

PhD Applicant | Public Health • Biostatistics • Applied Machine Learning • NLP • Data Science

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Public Health & Biostatistics | **Applied Machine Learning** | **Natural Language Processing (NLP)** | **Population & Humanitarian Research**

RESEARCH PROFILE

Statistics and Computer Science researcher with a dual master's background in Statistics and Computer Science & Engineering and 10+ years of combined experience across statistics, research, data analytics and information management. Research interests center on Public Health and Biostatistics, Applied Machine Learning and NLP, with particular interest in complex population data. Brings 3+ years of humanitarian information-management experience across 26 Rohingya refugee camps in Cox's Bazar, including management of approximately 4,500 protection cases in case-management workflows, design of three web-based information-management systems, and development of statistical, dashboard and data-collection workflows. Interested in doctoral research that combines rigorous statistical methods with practical computational approaches to population-health and resource-constrained data.

EDUCATION

MSc in Computer Science & Engineering | Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh | *Final result pending*

MSc in Statistics | Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh | *2022 · Focus: Data Mining, Text Mining, Biostatistics, Advanced Statistical Inference*

BSc (Hons) in Statistics | Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh | *2020*

HSC (Science) | Comilla Education Board, Bangladesh | *2015 · GPA 5.00/5.00*

SSC (Science) | Comilla Education Board, Bangladesh | *2013 · GPA 5.00/5.00*

RESEARCH EXPERIENCE & PROFESSIONAL EXPERIENCE

Senior Officer – Information Management | UNHCR – CODEC EPRC Project (Child Protection, GBV & Community-Based Protection)

January 2026 – June 2026 • Cox's Bazar, Bangladesh

- Managed protection programme data and analysis across 26 Rohingya refugee camps using Power BI, SQL, Python and Excel to support data quality, statistical analysis and evidence-based reporting.
- Led Power BI dashboard updates, data cleaning and analysis processes; reviewed and verified protection data prior to reporting.
- Conducted monthly protection-data analysis to identify trends, patterns and gaps across Child Protection, Gender-Based Violence and Community-Based Protection components.
- Maintained data-collection tools and information-management systems in line with programme needs and donor/sector guidance.

Information Management Manager (promoted from Information Management Specialist) | UNHCR – CODEC EPRC Project

January 2025 – December 2025 • Cox's Bazar, Bangladesh

- Oversaw information management for Child Protection, GBV and Community-Based Protection data across 26 refugee camps, ensuring data quality, compliance and actionable reporting.
- Managed information related to approximately 4,500 protection cases within case-management workflows, working with sensitive protection data and confidentiality requirements.
- Designed, deployed and maintained three web-based Information Management Systems supporting protection case data, indicator tracking and referral logs; reduced institutional reporting overhead by approximately 40%.
- Managed ActivityInfo (UNHCR) and CPIMS+ (UNICEF) datasets and served as project focal point for multi-sector reporting.
- Designed and maintained KoboToolbox/XLSForm tools with skip logic and validation rules; trained enumerators and field staff on data quality and security.
- Built Power BI dashboards and automated Excel/Python pipelines for KPI monitoring, beneficiary disaggregation and trend analysis.
- Organized five technical training sessions and contributed to capacity building for approximately 150 programme personnel.
- Applied data-protection and confidentiality practices to highly sensitive protection information.

Information Management Specialist | Community Development Centre (CODEC) – Education & Protection for Refugee Children (EPRC) Project

September 2024 – December 2024 • Cox's Bazar, Bangladesh

- Supported design, deployment and maintenance of information-management systems covering 26 Rohingya refugee camps.
- Assisted with ActivityInfo (UNHCR) and CPIMS+ (UNICEF) data entry, cleaning, validation and indicator-wise reporting.
- Developed KoboToolbox, MS Forms and Google Forms data-collection tools and conducted preprocessing and statistical analysis for programme monitoring.
- Contributed to Power BI/Excel dashboards and coordinated with Case Management and M&E teams to maintain accurate and compliant information flows.

Research Intern — Statistics & Data Science | Leading University, Department of Computer Science & Engineering

June 2022 – June 2024 • Sylhet, Bangladesh

- Conducted research on digital privacy and security for children using statistical modelling, hypothesis testing, survey design and sampling.
- Applied machine learning and GIS-based spatial analysis; developed dashboards and visualizations using Power BI, Tableau and Google Data Studio.
- Completed 30+ analytical practice projects and contributed to 20+ professional research projects, including budget-monitoring work.

Information Management Officer | Society for Sylhet Resources Advancement Community (SRAC)
February 2020 – May 2022 • Sylhet, Bangladesh

- Managed digital information systems and databases for government- and SRAC-funded skill-development programmes, overseeing the full data lifecycle of trainee records.
- Prepared Monthly Progress Reports in line with government and donor reporting standards.
- Led the transition from paper-based to digital data collection during COVID-19 to support programme continuity.
- Maintained monitoring tools for outputs, outcomes and indicators and supported audit/compliance requirements.

Mathematics, Physics & ICT Teacher (Part-Time) | Focus University Admission Coaching / Radium Engineering & University Admission Coaching
February 2016 – May 2022 • Sylhet, Bangladesh

- Taught higher mathematics, probability, statistics fundamentals and quantitative problem-solving.
- Applied performance analytics and data-driven evaluation methods to improve learning outcomes.

SELECTED RESEARCH PROJECTS

Bengali Math-Quest NLP System — Deep-learning question-answering system for Bengali-language mathematics; Python, TensorFlow.

Spatial Analysis of COVID-19 Spread Across Bangladesh — District-level GIS mapping and inferential statistics to identify high-risk transmission areas; ArcGIS, R.

Rainfall Prediction for Sylhet District — SARIMA time-series forecasting model; R.

Digital Privacy & Security for Children — Statistical modelling, hypothesis testing, survey design and sampling, with machine learning and GIS-based analysis.

PUBLICATION

Sarker, N., Sorkar, M. N. U., Sorkar, M. R. U., & Uddin, M. T. (2022). Forecast, Trend, Growth, and Instability of Internet Users in Bangladesh. Computer Engineering and Intelligent Systems, 13(4). DOI: 10.7176/CEIS/13-4-01

RESEARCH & TECHNICAL SKILLS

Statistical & Research Methods: Descriptive and inferential statistics; statistical modelling; hypothesis testing; survey/assessment design; sampling; biostatistics; time-series analysis.

Machine Learning & NLP: Applied machine learning; deep learning; Bengali NLP; TensorFlow; Python; R.

Data & Analytics: SQL; MySQL; Power BI (DAX, Power Query); R Shiny; Excel; Tableau; Google Data Studio.

GIS & Spatial Analysis: ArcGIS; QGIS; GIS-based analysis and visualization.

Information Systems: KoboToolbox/XLSForms; ActivityInfo (UNHCR); CPIMS+ (UNICEF); data collection and validation workflows.

Data Quality & Governance: Cleaning; validation; deduplication; cross-variable checks; data-sharing protocols; confidentiality and responsible-data practices.

Other: Git/GitHub; Microsoft Office; PowerPoint; Visio.

TRAINING & PROFESSIONAL DEVELOPMENT

- University Innovation Hub Program (UIHP – SUST Cohort-1), Bangladesh High Tech Park Authority — 6 months, 2024.
- 60-Hour Training Course on Research Techniques in Social Sciences, SUST Research Centre / Social Science Research Council, Ministry of Planning — 2022.
- Online professional courses: Python & Django Framework, JavaScript Web Development, AI/Text-to-Image Tools (Udemy).

SELECTED RESEARCH DIRECTION

Potential doctoral directions: biostatistical and population-health modeling; applied machine learning for health and population data; NLP for low-resource and domain-specific applications; and responsible analysis of sensitive population data. These interests build on existing foundations while leaving the specific doctoral question and methodology open to the supervisor, research group and scientific problem.

LANGUAGES

Bengali — fluent (written and spoken) · English — fluent (written and spoken)

REFERENCES

Akter Hossain — MEAL Manager, UNHCR – CODEC EPRC Project, Cox's Bazar; Former Supervisor; mealcp.eprc@codec.org.bd

Rumel M. Saifur Rahman Pir — PhD Researcher & Senior Teaching Assistant, Swansea University, Wales, UK; Former Research Supervisor; rumelpir@gmail.com

Mohammad Abdullah Al Mumin, PhD — Professor, Department of Computer Science & Engineering, Shahjalal University of Science & Technology (SUST); Academic Supervisor; mumin-cse@sust.edu